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LATER STAGE LARVAE OF *PANULIRUS GUTTATUS* (LATREILLE, 1804) (DECAPODA, PALINURIDAE) WITH NOTES ON THE IDENTIFICATION OF PHYLLOSOMATA OF *PANULIRUS* IN THE CARIBBEAN SEA

BY

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ABSTRACT

Larval stages VI to X of *Panulirus guttatus* from Cuban waters are described. The principal differences with the larvae of the other two Caribbean species of *Panulirus* are discussed.

RÉSUMÉ

Les caractères des larves VI à X de *Panulirus guttatus* des eaux proches de Cuba sont décrites. Les principales différences entre les larves des trois espèces de *Panulirus* signalées dans la région caraïbe sont également exposées.

INTRODUCTION

The Caribbean spiny lobster *Panulirus argus* (Latreille, 1804), the Guinea lobster *P. guttatus* (Latreille, 1804) and the green or smooth-tailed lobster *P. laevicauda* (Latreille, 1804) are the only species of *Panulirus* reported for the Caribbean area (Holthuis & Zaneveld, 1958). One of the main difficulties faced during larval ecological studies of the commercial species of spiny lobsters is the possibility of confusing the larvae of closely related species. While phyllosomata of *P. argus* are very well known (Crawford & De Smidt, 1922; Lebour, 1950; Lewis, 1951; Baisre, 1964) and two later larvae assigned to *P. laevicauda* have been already described by Baisre & Ruiz de Quevedo (1982), there is not a published description on larvae of *P. guttatus* although Robertson (1968) mentioned to have obtained the first larvae of this species from ovigerous females hatched in the laboratory.

When discussing larval groups within *Panulirus*, Baisre & Ruiz de Quevedo (1982) have shown that larvae of *P. argus* belong to a group very different from that of *P. laevicauda*. According to this paper, larvae of *P. guttatus* whose adults belong to the same group as *P. argus*, must share the same morphological features which distinguish phyllosoma larvae of *P. argus*, from those of *P. laevicauda*. During the routine examination of plankton samples from oceanic waters around Cuba, searching for *P. argus* larvae, we have found 48 phyllosoma larvae that undoubtedly belong to *Panulirus* and which are clearly neither *P. argus* nor *P. laevicauda*. After a close examination of these larvae and by comparison with other phyllosomata of *Panulirus* we conclude that there is strong evidence for assigning them to *P. guttatus*.

In this paper, a description of these larvae is presented together with a discussion of the larval characters of *P. guttatus*. The main differences between the three species of *Panulirus* reported for the Caribbean area are also included.

MATERIAL AND METHODS

The pyllosoma specimens described in this paper were obtained incidentally during a sampling program designed for the assessment of distribution and larval ecology of the commercial species of spiny lobster *P. argus* and most of the samples come from the cruises carried out by the R/V "Ulises" in waters around Cuba. Date and location where specimens of *P. guttatus* were collected are summarized in table I. Drawings were made with a Wild binocular stereomicroscope with a drawing tube and all the morphological features of the larvae which are mentioned in the text are shown in fig. 1, which represents a hypothetical final stage of *Panulirus*. Measurements were made using an ocular micrometer. The total length was measured from the anterior margin of the cephalic shield between the eyestalks to the posterior margin of the telson; width of the cephalic shield and width of the thorax were taken across the widest part of both. The distance from the midpoint between the coxal segments of the second maxilliped (a), to the mouth parts (b) (labrum and first maxilla) is expressed as a proportion (b:a) of the distance between the attachment of the coxae to the thorax as defined by Johnson (1968). Similar measurements on specimens of *P. argus* and *P. laevicauda* were also made for comparison.

DESCRIPTIONS

Panulirus guttatus (Latreille, 1804)

Phyllosoma stages

Stage VI (fig. 2). Length 11.0-14.9 mm (13 specimens). The cephalic shield is narrower than the thorax, the ratio being 0.80-0.83. Eyestalk 3.2-4.2 mm. Antennule longer than the antenna. Dorsal and ventral coxal spines and subexopodal spines absent. Fifth pair of pereopods visible as small buds. Abdomen with traces of segmentation and uropods visible as small buds. Pleopods absent.

Stage VII (fig. 3). Length 15.3-19.5 mm (22 specimens). Cephalic shield width : thorax width ratio 0.84-0.85. Eyestalk 4.4-5.7 mm. Antennule still longer than the antenna. The distal segment of the maxilla (scaphognathite) is now expanded and covered by setae. The exopod of the second maxilliped is now visible as a small bud (fig. 3b). The pleopods are also visible as small buds and the abdomen is fully segmented; uropods biramous but unsegmented (fig. 3c).

TABLE I

Date and location where phyllosomata of *Panulirus guttatus* were collected

Date	Lat. N	Long. W	Number of larvae	Stages	Gear
Aug./1979	20°00′	78°00′	1	VII	Trap
	18°55′	75°20′	1	VII	Trap
	19°46′	76°42′	1	VII	Trap
Apr./1980	20°24′	78°30′	1	VII	Neus
Oct./1980	19°40′	75°25′	1	VIII	Neus
Jun./1986	21°25′	83°00′	1	IX	Trap
	21°30′	76°00′	3	VIII	Neus
Aug./1986	19°30′	78°00′	1	IX	Trap
	19°50′	75°00′	1	X	Trap
	19°40′	76°00′	3	VI	Trap
	”	”	1	VII	Trap
	”	”	1	VIII	Trap
Oct./1986	21°30′	80°42′	1	VII	Trap
	21°25′	80°55′	1	VII	Neus
	20°00′	79°00′	8	VI	Neus
	”	”	5	VII	Neus
	”	”	1	X	Neus
Mar./1987	21°30′	80°00′	1	IX	Trap
Jul./1989	21°40′	80°36′	1	IX	Trap
	21°30′	79°40′	4	VII	Trap
	21°40′	80°40′	3	VII	Trap
Aug./1989	19°36′	76°00′	3	VII	Trap
	20°10′	78°35′	1	VIII	Trap
	20°25′	79°35′	2	VI	Trap
	20°40′	79°25′	1	VIII	Trap

Trap = Trapecio Net (Guitart, 1971).
Neus = Neuston Net.

Stage VIII (fig. 4). Length 19.4-21.0 mm (7 specimens). Cephalic shield width : thorax width ratio 0.84-0.85. Eyestalk 5.6-6.8 mm. The antenna is now as long as the antenna (fig. 4b). Anterior branch (basis) of the maxillule with a rudimentary endopod (palp) indicated by two setae on a small protuberance and three strong distal spines (fig. 4d). Scaphognathite of the maxilla more expanded than in the previous stage and fringed by setae (fig. 4c). As in other species of *Panulirus*, the dactyl of the second pereiopod is large and prehensile while those of the first, third and fourth pereiopods are short (fig. 4e-h). The fifth pair of pereiopods reaches almost half the length of the abdomen, but is still unsegmented. Uropods segmented.

Stage IX (fig. 5). Length 22.5-29.8 mm (4 specimens). Cephalic shield width : thorax width ratio 0.83-0.89. Eyestalk 7.1-8.1 mm. The antenna is now longer than the antennule and the eyestalk, and has a spatulate tip. The fifth pair of

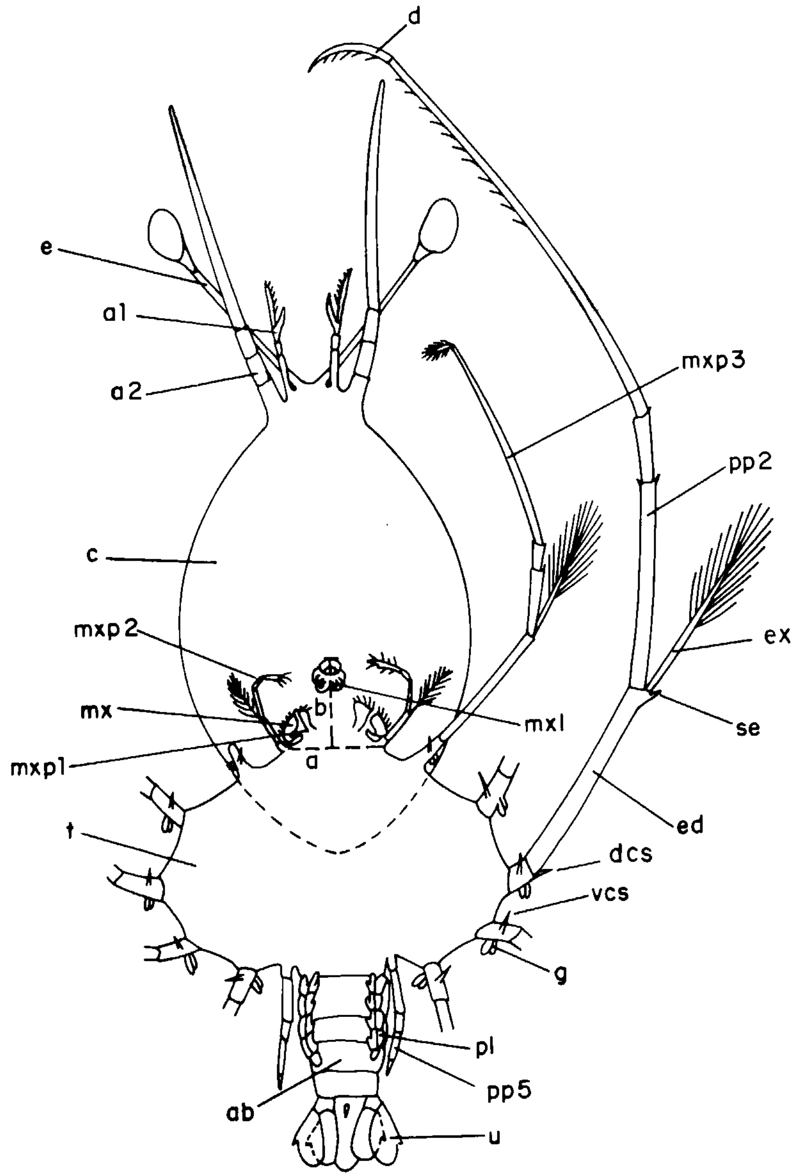

Fig. 1. A final stage phyllosoma of *Panulirus* labelled to indicate morphological features which are mentioned in text. a1, antennule; a2, antenna; ab, abdomen; d, dactyl; dcs, dorsal coxal spine; e, eyestalk; ed, endopod; ex, exopod; g, gills; mxl, maxillule; mx, maxilla; mxp1, first maxilliped; mxp2, second maxilliped; mxp3, third maxilliped; p1, pleopod; pp2, second pereiopod; pp5, fifth pereiopod; t, thorax; u, uropod. The distances a and b are defined in the text.

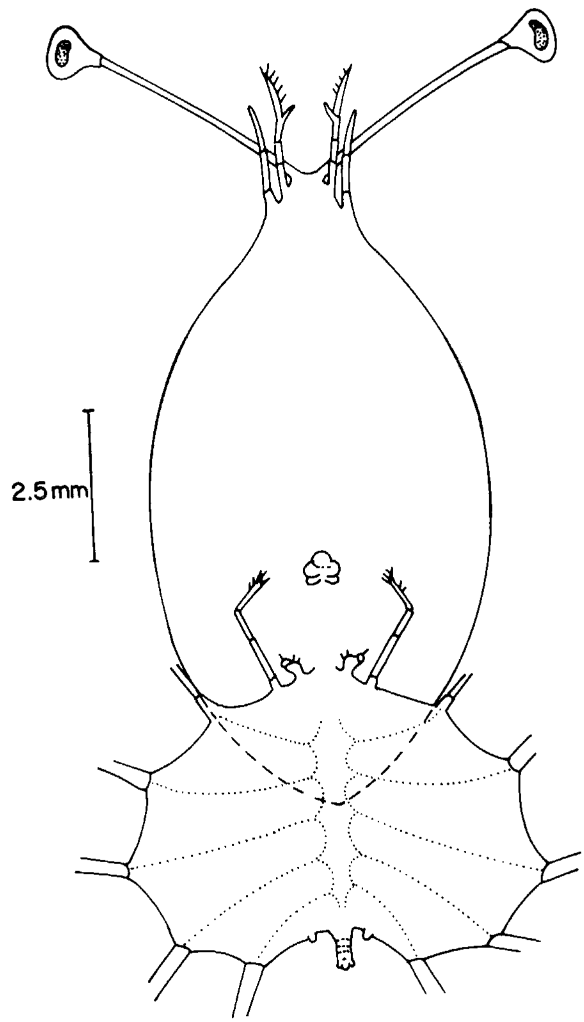

Fig. 2. *Panulirus guttatus* Latreille, stage VI phyllosoma, ventral view.

pereiopods has three segments. The pleopods are biramous and the uropods are fully formed.

Stage X (fig. 6). Length 31.9-34.9 mm (2 specimens). Cephalic shield width : thorax width ratio 0.83-0.92 mm. The spatulate tip of the antenna is now easily noticeable (fig. 6b). The exopod of the second maxilliped is now provided with four setae and the first maxilliped nearly reaches the edge of the scaphognathite (fig. 6c). The fifth pereiopod has five segments. The pleopods have a small appendix interna and there is a small spine on the external margin of each uropod. The bilobed gills, characteristic of this stage in most species of *Pan-*

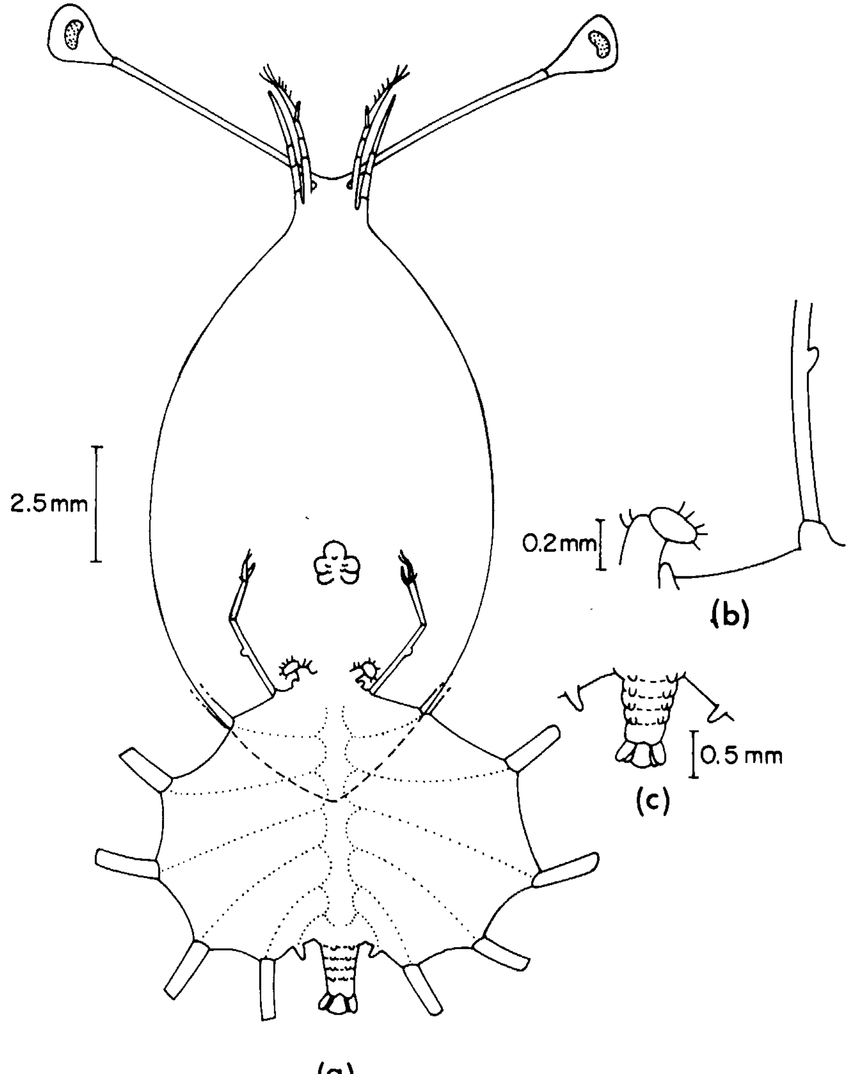

Fig. 3. *Panulirus guttatus* Latreille, stage VII phyllosoma. a, vental view; b, maxilla and second maxilliped; c, abdomen.

ulirus, appear only as small papillae on the dorsal side of the coxa. This feature has also been reported for *Panulirus penicillatus* (Olivier, 1791) and *P. marginatus* (Quoy & Gaimard, 1825) by Johnson (1968).

IDENTIFICATION OF THE LARVAE

Basic morphometrics of the all three species of *Panulirus* are given table II, while prominent differences between these species are presented in table III.

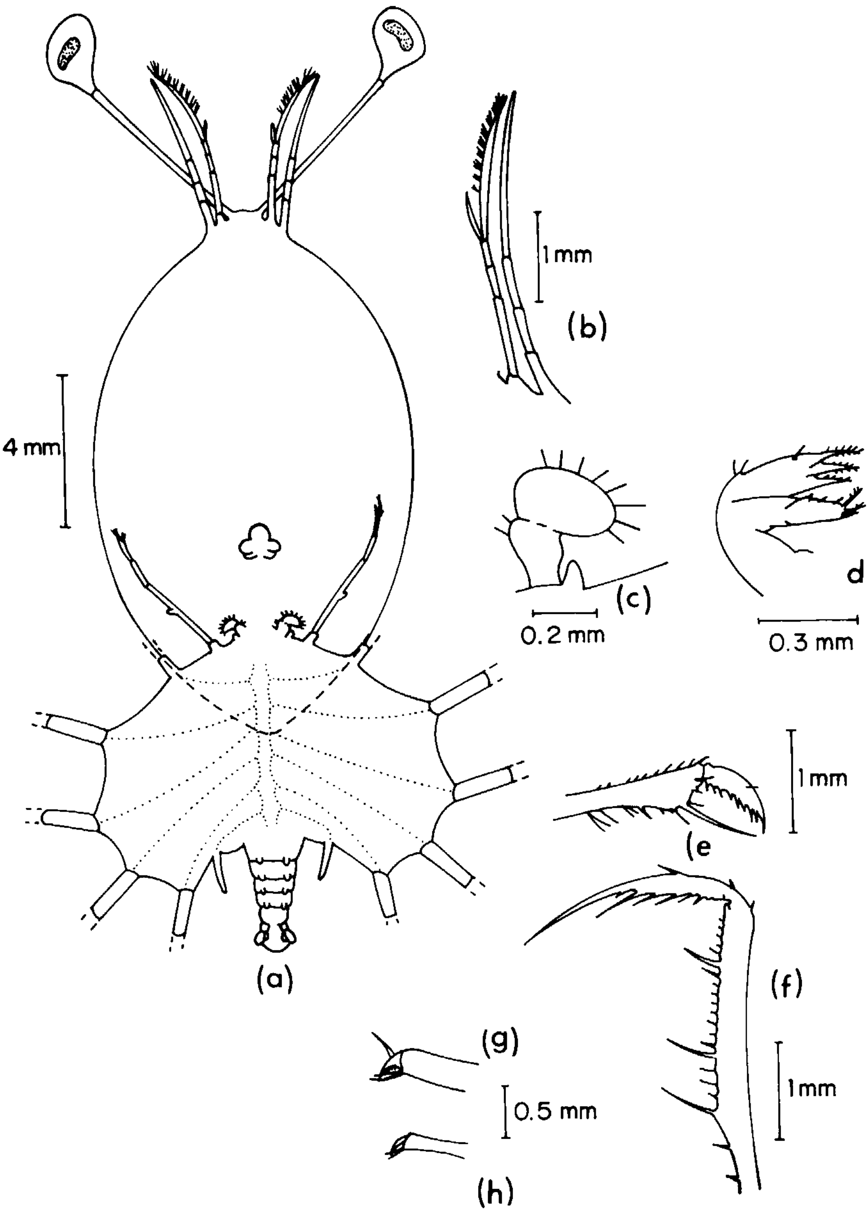

Fig. 4. *Panulirus guttatus* Latreille, stage VIII phyllosoma. a, ventral view; b, antenna and antennule; c, maxilla and first maxilliped; d, maxillule; e, dactyl of the first pereiopod; f, second pereiopod; g, third pereiopod; h, fourth pereiopod.

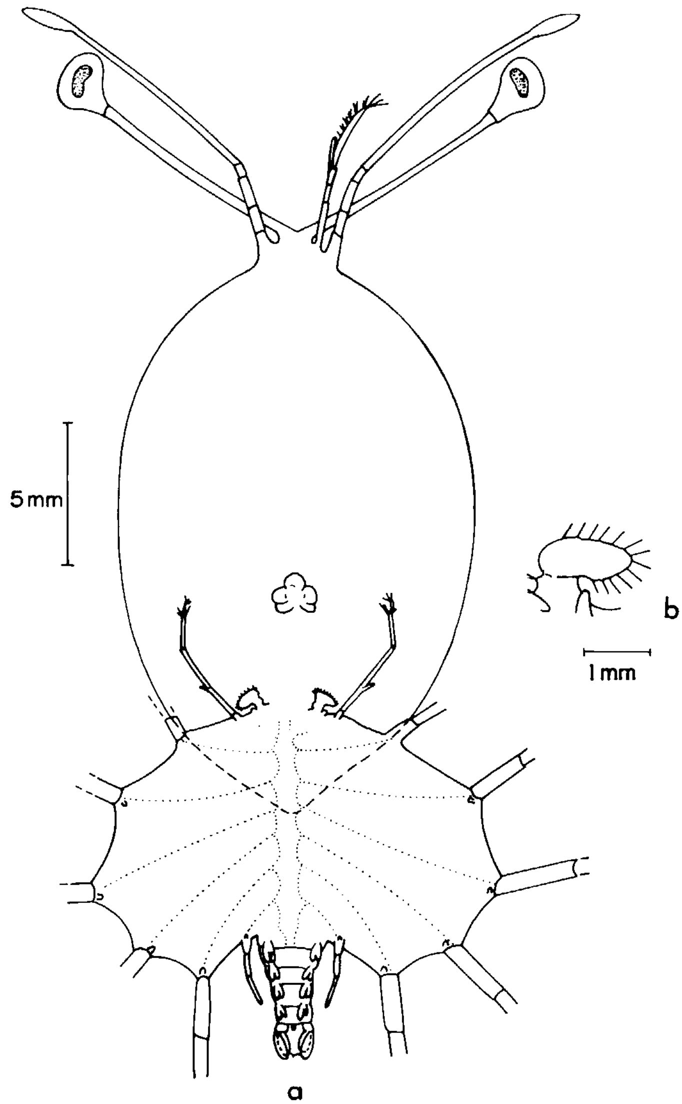

Fig. 5. *Panulirus guttatus* Latreille, stage IX phyllosoma. a, ventral view; b, maxilla and first maxilliped.

Although larvae described in this paper clearly belong to *Panulirus* they are quite different from all the other phyllosomata of that genus reported for the Caribbean area. The major distinguishing features of later stage larvae assigned to *P. guttatus* are:

a) Larger size as compared to those of *P. argus* and *P. laevicauda*.

b) The ratio cephalic shield width : thorax width (0.8-0.9) is intermediate between that of *P. argus* (less than 0.8) and of *P. laevicauda* (close to 1.0).

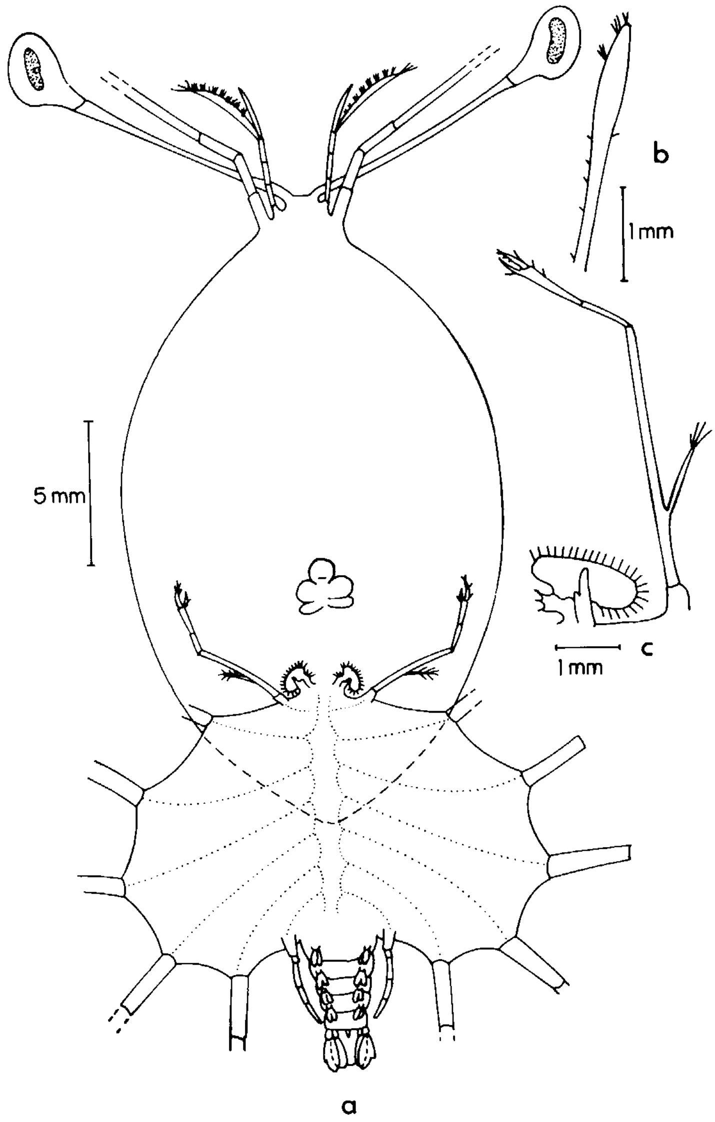

Fig. 6. *Panulirus guttatus* Latreille, stage I phyllosoma. a, ventral view; b, spatulate tip of the antenna; c, maxilla, first and second maxilliped.

c) Ratio a : b (1.3-1.6) larger than in *P. argus* (1.1-1.3) and in *P. laevicauda* (less than 1.0).

d) Presence of a spatulate tip in the second antenna in stages IX and X.

e) The longer eyestalk.

Apart from these differences, larvae of *P. guttatus* share several features with larvae of *P. argus*, like the absence of subexopodal spines, the absence of ventral

TABLE II

Basic morphometrics of larvae assigned to *Panulirus guttatus* (range, mean size -in mm- and SE±95% confidence limits -in mm- are included)

Stage		Total length	Cephalic shield length	Cephalic shield width	Thorax width	Eyestalk length	Abdomen length	b/a[1]	Width to length of cephalic shield
VI	range	11.0-14.9	8.1-11.5	4.4- 6.7	5.7- 7.7	3.4-4.2	0.7-0.9	1.1-1.7	
n = 13	x̄	13.42	10.42	5.72	7.02	3.88	0.8	1.41	0.67
	SE	0.67	0.57	0.35	0.31	0.16	0.04	0.12	
VII	range	15.3-19.5	13.0-14.0	6.7- 9.2	8.3-10.3	4.4-5.7	1.0-1.6	1.0-1.7	
n = 22	x̄	17.13	13.53	7.67	9.05	5.2	1.19	1.42	0.67
	SE	0.37	0.16	0.24	0.24	0.06	0.06	0.08	
VIII	range	19.4-21.0	14.7-16.3	8.6- 9.1	10.3-11.0	5.6-6.8	1.3-1.9	1.3-1.5	
n = 7	x̄	20.11	15.31	8.87	10.53	6.07	1.56	1.43	0.69
	SE	0.39	0.39	0.18	0.18	0.27	0.14	0.08	
IX	range	22.5-29.8	17.1-20.8	10.0-12.5	11.9-14.1	7.1-8.1	2.0-5.3	1.3-1.6	
n = 4	x̄	25.2	18.5	11.0	12.8	7.5	3.15	1.5	0.69
	SE	3.1	1.6	1.1	0.82	0.47	1.4	0.12	
x̄	range	31.9-34.9	22.8-24.1	13.6-14.0	15.3-16.9	3.5-3.7	4.7-5.4	1.5	
n = 2	x̄	33.4	23.45	13.8	16.1	3.6	5.05	1.5	0.69
	SE	2.9	1.27	0.39	1.6	0.2	0.7	0	

(1) As defined by Johnson (1968). See also fig. 1.

TABLE III

Prominent differences between phyllosomata of the three Caribbean species of *Panulirus* (n = number of observations; TL = total length; CS = cephalic shield; T = thorax; VCS = ventral coxal spines; SS = subexopodal spines; EY = eyestalk; MX = second maxilla; ab = absent; pr = present; MXP = maxilliped)

Stage	Species	n	TL (mm)	CS width : T width	VCS	SS	b / a	EYlength (mm)	Setae in MX2	Tip of the antennae
	1. *P. argus*	30	8.65	0.77	ab	ab	1.29	2.27	ab	pointed
VI	2. *P. guttatus*	13	13.4	0.81	ab	ab	3.84	3.84	pr	pointed
	3. *P. laevicauda*	–	–	–	–	–	–	–	–	–
	1. *P. argus*	13	10.85	0.79	ab	ab	1.29	3.03	ab	pointed
VII	2. *P. guttatus*	22	17.13	0.84	ab	ab	1.42	5.02	pr	pointed
	3. *P. laevicauda*	1	11.2	0.98	MXP3 = P1	P1-P4	1.0	2.90	pr	pointed
	1. *P. argus*	27	13.98	0.79	ab	ab	1.23	3.78	pr	pointed
VIII	2. *P. guttatus*	7	20.2	0.84	ab	ab	1.38	6.18	pr	pointed
	3. *P. laevicauda*	–	–	–	–	–	–	–	–	–
	1. *P. argus*	4	17.99	0.80	ab	ab	1.17	4.72	pr	pointed
IX	2. *P. guttatus*	4	25.4	0.90	ab	ab	1.45	7.40	pr	spatulated
	3. *P. laevicauda*	1	17.5	1.01	MXP3 = P1	P1-P4	1.0	5.20	pr	pointed
	1. *P. argus*	4	21.5	0.78	ab	ab	1.10	5.38	pr	pointed
X	2. *P. guttatus*	2	33.4	0.90	ab	ab	1.40	8.75	pr	spatulated
	3. *P. laevicauda*		–	–	–	–	–	–	–	–

coxal spines in the third maxilliped and first pereipod and the absence of dorsal coxal spines. All of these features are present in larvae of *P. laevicauda.*

As *P. guttatus* is a common species in the Caribbean area and the only species whose larvae have not been described and taking into account that the larvae here described are relatively abundant and show the closest resemblance to those of *P. penicillatus* and *P. marginatus*; two Indopacific species closely related to *P. guttatus* (see Holthuis, 1961), we are safe to assume that they belong to this species. Although *P. echinatus* Smith, 1869, the fourth Atlantic species of *Panulirus,* is also closely related to *P. guttatus* (see Holthuis, 1961), we have not considered the possibility that larvae described in this paper could be assigned to that species. In the Western Atlantic area, *P.echinatus* is restricted to the Brazilian east coast between 8°N, to 23° S (Vianna, 1986) and the possibility of its larvae to be transported to the Caribbean area can be ruled out.

DISCUSSION

Although closer to *P. argus* than to *P. laevicauda,* larvae reported in this paper are clearly distinguishable from those of *P. argus* by the larger size, the distance between the mouth-parts, the ratio between cephalic shield width and thorax width, the long eyestalk and the spatulate tip of the antenna in later stages. The presence of a spatulate antenna is, however, most unexpected. To the author's knowledge, only the last phyllosoma of *Panulirus regius* De Brito Capello, 1864 (see Crosnier, 1971, fig. 9c) and of *P. homarus* (L., 1758) (see Michel, 1971), and the puerulus of *P. versicolor* (Latreille, 1804) (see Gordon, 1953; Berry, 1974) and *P. ornatus* (Fabricius, 1798) (see Berry, 1974) have this feature. According to Baisre & Ruiz de Quevedo (1982) all of these species belong to a group quite different from that of *P. argus* and *P. guttatus.* The closest resemblance of larvae described in this paper are with larvae of *P. penicillatus* and *P. marginatus,* from the Indo-West Pacific area, and none of these two species have such an antenna. However, since antennae are fragile it is often difficult to determine with certainty whether or not the tips are widened in a spatulate way because many tips appear to have been broken-off in animals collected with plankton nets.

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